

1. Prove parts (v) and (vi) of the Correspondence Theorem: Let G be a group and $K \triangleleft G$. Define a map

$$\Theta : \{H : H \text{ is a subgroup of } G, K \subseteq H\} \rightarrow \{\mathcal{H} : \mathcal{H} \text{ is a subgroup of } G/K\}$$

by $\Theta(H) = H/K$. If H_1 and H_2 are subgroups of G containing K , then

(v) Θ preserves intersections: $(H_1/K) \cap (H_2/K) = (H_1 \cap H_2)/K$,

(vi) Θ preserves products: if $H_1 H_2 \leq G$ then $(H_1/K)(H_2/K) = (H_1 H_2)/K$.

2. Consider the group $G = \mathbb{Z}_{36}$ and subgroups $H = \langle [9] \rangle$ and $K = \langle [18] \rangle$.

(a) Determine all subgroups of $\mathbb{Z}_{36}$ and draw the lattice diagram.

(b) Use the Correspondence Theorem to draw the lattice diagrams for $\mathbb{Z}_{36}/H$ and $\mathbb{Z}_{36}/K$.

3. Repeat Question 2 with $G = \mathbb{Z}_{100}$, $H = \langle [20] \rangle$ and $K = \langle [25] \rangle$.

4. Let H , K and L be normal subgroups of G with $H \subseteq K \subseteq L$. Define $A = G/H$, $B = K/H$ and $C = L/H$.

(a) Show that $B \triangleleft A$, $C \triangleleft A$ and $B \triangleleft C$.

(b) What factor group is $(A/B)/(C/B)$ isomorphic to?

5. (a) Prove that a finite group with only two subgroups (itself and the trivial group) is cyclic and of prime order.

(b) Let G be a finite group with $K \triangleleft G$. Show that the following conditions are equivalent:

(i) If H is a subgroup of G with $K \subseteq H$, then $H = K$ or $H = G$,

(ii) G/K is cyclic and has prime order.

Extra problems

A normal subgroup K of G (where $K \neq G$) is called a **maximal normal subgroup** if the only normal subgroups H of G with $K \subseteq H$ are $H = K$ and $H = G$. A group G is called **simple** if its only normal subgroups are $\{1\}$ and G .

-
- (I) Prove that if G is simple and $\varphi : G \rightarrow G'$ is an epimorphism, then G' is simple.
- (II) Prove that a normal subgroup K of G is a maximal normal subgroup if and only if G/K is simple.
- (III) Let $K \triangleleft G$ be a maximal normal subgroup of G and $H \triangleleft G$ where $K \not\subseteq H$.
- (i) Justify why $G = HK$.
- HINT: Show $HK \triangleleft G$.
- (ii) Show that $H/(H \cap K)$ is simple.